Residual Penalties Versus Algebraic Completion In Conditional Diffusion

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Abstract

In solar wind imputation, a variance identity restricts possible combinations of magnetic statistics. Conditional diffusion must learn their probabilities given available measurements. The energy score compares predictive distributions: a penalty on the identity residual gives little mean improvement over diffusion alone when validation chooses the model. The penalty improves the score at the end of each training run. Generating the quantities needed to derive the remaining statistic gives the lowest mean score on those quantities. Both representation and training loss change, and prediction intervals cover fewer withheld values than intended. Clipping can enforce moment constraints and worsen the score through reduced ensemble spread. An exact denoiser example gives small residuals with inaccurate conditional samples. Consistency and distributional accuracy need separate assessment. Evaluation withholds parts of complete records from a test period examined during earlier development.

1 Introduction

Missing solar wind measurements complicate estimates of magnetic turbulence [1]. Changes in field direction can produce a small mean vector despite large individual magnitudes. Interpolating the mean alone leaves the magnetic fluctuations undetermined. The models infer the joint distribution of six magnetic statistics over a missing hour from measurements before and after it.[1]

A variance identity determines one statistic from the other five, restricting their possible combinations while leaving their conditional probabilities to be learned. It holds for arbitrary vectors, including noise, and is independent of plasma dynamics. The models use conditional diffusion [2] with the residual objective and estimator of Physics-Informed Diffusion Models [3]. Earlier work compares residual minimization with distribution matching [4] and enforces algebraic constraints through output construction [5]. Comparing a residual penalty with algebraic completion and clipping tests whether satisfying the constraints improves prediction. The penalty's apparent benefit depends on which model state is selected from training.

2 Magnetic statistics and data

Let B_j denote five successive OMNI vector averages, each over one minute, from the same spacecraft in geocentric solar ecliptic coordinates. Define

$$\mu = \frac{1}{5}\sum_j B_j, \quad m = \frac{1}{5}\sum_j \|B_j\|, \quad v_s = \frac{1}{5}\sum_j (\|B_j\| - m)^2, \quad v_v = \frac{1}{5}\sum_j \|B_j - \mu\|^2. \quad (1)$$

[1]Available at: https://github.com/magnaprog/diffusion-consistency

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The six variables are $z = (\mu_x, \mu_y, \mu_z, m, v_s, v_v)$. Expanding both population variances gives

$$R_{\text{obs}}(z) = m^2 + v_s - \|\mu\|^2 - v_v = 0. \tag{2}$$

Omitting the variances gives $R_{\text{inst}} = m^2 - \|\mu\|^2$. A penalty on this quantity suppresses cancellation between field directions. It vanishes when all nonzero vectors share a direction, even if their magnitudes differ.

Write $r = \|\mu\|$. Five vectors in three dimensions, including possible zero vectors, can realize these statistics exactly when

$$m \geq r, \qquad 0 \leq v_s \leq \frac{3}{2}m^2 + \frac{5}{2}r^2, \qquad v_v = m^2 + v_s - r^2. \tag{3}$$

Appendix A proves sufficiency. Algebraic completion changes only v_v, so the primary score over the other five variables is unchanged. Clipping restricts m and v_s to the ranges in Equation 3, then derives v_v. Changes to m or v_s can affect the score.

The conditional mean can lie outside the nonlinear set of possible statistics even when every sample obeys the identity. For conditioning information Y,

$$R_{\text{obs}}(\mathbb{E}[z \mid Y]) = \operatorname{tr} \operatorname{Cov}(\mu \mid Y) - \operatorname{Var}(m \mid Y). \tag{4}$$

Either sign is possible. A residual at a denoiser's mean prediction can therefore reflect conditional uncertainty. Residuals at individual samples test whether those samples obey the identity [3].

Data construction. The data come from the modified OMNI high resolution product [6, 7]. Timestamps estimate arrival at the bow shock nose; propagation and averaging precede masking [8]. The task reconstructs records on this time grid; recovery at the spacecraft's original observation times requires a separate experiment. Magnitudes are Euclidean norms of the vector components. Each window spans four hours in 48 bins of five minutes. The central twelve bins are masked without prior imputation, leaving 90 minutes of conditioning data on each side. Complete windows use one spacecraft within one month, retaining 12.77% of possible training starts.

In every year from 2005 through 2024, training uses 15 January through 30 June, test uses August and September, and validation uses 1 November through 17 December. The dates between these periods separate the sets. Normalization uses the 300,392 distinct bins in 122,309 overlapping training windows. Evaluation first packs windows of four hours without overlap, then selects evenly spaced positions within each month up to specified quotas, giving 319 validation and 480 test windows.

3 Models and evaluation

Training. D uses diffusion to generate all six variables [9]. I-S adds a penalty on R_{inst}; O-S adds one on the variance identity R_{obs}. G5 generates five variables and derives v_v after sampling. All four models receive all six variables in the available intervals. Training and sampling add noise only within the missing hour and preserve the available measurements.

For standardized state u in the missing hour, the training objective adapts Bastek et al. [3] by combining mean squared error with Min-SNR-5 weighting [10] and a residual term:

$$\mathcal{L}_t = w_t \ \text{MSE}(\hat{u}_0, u_0) + \frac{c_R}{2\Sigma_t} \frac{1}{12} \sum_{k=1}^{12} \left(\frac{R(z^*_{0,k})}{B^2_{\text{ref}}} \right)^2. \tag{5}$$

Here $\hat{u}_0$ estimates the signal before noise. Two denoiser evaluations give $z^*_{0,k}$ in physical units for bin k, except at $t = 1$, where the first estimate is reused. The weight is $w_t = \min(\text{SNR}_t, 5)$, with signal-to-noise ratio SNR_t; B^2_{ref} is the mean of $m^2 + v_s$ over distinct training bins. The cosine noise schedule has 100 steps [11]. Σ_t is the posterior variance, with $\Sigma_1 := \Sigma_2$. D and G5 use $c_R = 0$; I-S and O-S use $c_R = 0.0005$.

Each method uses three training seeds and 29,636 parameter updates, saving five model states, or checkpoints, per seed (Appendix C). Residual estimation uses extra denoiser evaluations, so equal update counts involve different amounts of computation. The comparison uses the original settings and penalty coefficient without further tuning on the test period examined earlier.

Scores and uncertainty. The energy score evaluates predictive distributions using Euclidean distances [12]; lower scores are better. The primary score uses $(\mu_x, \mu_y, \mu_z, m, v_s)$ at twelve hidden positions, giving 60 coordinates per window standardized using training data. For K draws, its unbiased estimate is the mean distance to withheld values minus half the mean distance over the $K(K-1)$ ordered pairs of distinct draws. Scores give equal weight to each month in the evaluation. Other measures are the continuous ranked probability score, coverage and interval width; Appendix C gives their definitions. Diffusion results average three training seeds. Coverage is computed separately in each ensemble of 39 draws before averaging. Intervals for score differences come from resampling years, or groups of two consecutive years, with replacement. Each resampling replicate uses the same years for all models and retains the month counts for each occurrence of a year. These intervals describe variation across test years for the existing model fits and draws. Grouping windows by withheld magnitude or increments gives retrospective comparisons of those test cases.

Model selection and baselines. The 60 diffusion validation ensembles each contain 39 draws, using 25 denoising diffusion implicit model (DDIM) evaluations per draw with $\eta = 0$ [13]. Validation selects checkpoints separately for raw and clipped predictions by minimizing the energy score with equal month weights; exact ties favor the earlier state. Test ensembles contain 39 draws with 50 evaluations each.

Linear interpolation joins the available bins immediately before and after the missing hour; persistence repeats the preceding bin. Gaussian interpolation adds a Gaussian residual vector with 72 coordinates to the interpolant, using a mean and covariance from TRAIN and the original covariance shrinkage $\rho = 0$. Gaussian regression predicts the 60 primary residuals from all 216 input coordinates, uses a constant covariance, and derives v_v. Validation chooses its ridge regularization and covariance shrinkage parameters before TEST (Appendix G). All comparisons reuse the same baseline ensembles.

3.1 Test results

O-S and D have close mean energy scores (Table 1). O-S minus D is -0.0028, with a 95% interval of $[-0.0346, 0.0242]$ from resampling years. Relative intervals from both resampling schemes fall within the original $\pm 2\%$ comparison range for these three fits. G5 has the lowest mean score, although nearly half of its raw bins fail the moment inequalities and its predictive intervals provide less than their nominal coverage. O-S minus G5 is 0.1002, with interval $[0.0712, 0.1391]$ from resampling years. Both comparisons change sign across seeds. Greater ensemble spread accounts for most of G5's mean score improvement over D, with contributions that differ by seed (Figure 4). G5 changes both generation dimension and training loss; deriving v_v alone leaves the primary score unchanged.

Table 1: Results for 480 test windows with equal month weights. Energy scores use the 60 standardized primary coordinates. Diffusion results average three fits; each other baseline uses one ensemble realization. Validation selects checkpoints separately for raw and clipped diffusion predictions. Coverage is averaged over five primary variables at twelve hidden positions. $|R|$ is mean absolute R_{obs} in nT^2. The last column gives the percentage of bins that fail a moment inequality beyond rounding, excluding the equality test. Interpolation and persistence have intervals of zero width.

	Energy score		90% coverage (%)		Raw samples	
Procedure	Raw	Clipped	Raw	Clipped	$\lvert R\rvert$	Bins failing inequalities (%)
D	1.888	1.869	72.8	71.0	2.956	50.4
I-S	1.882	1.876	69.4	69.6	3.732	37.8
O-S	1.885	1.866	70.0	71.1	2.680	55.7
G5	1.785	1.784	75.7	75.8	0.000	48.8
Interpolation	2.393	2.393	0.0	0.0	3.459	0.0
Persistence	3.067	3.067	0.1	0.1	0.000	0.0
Gaussian interpolation	2.125	2.229	94.3	93.6	15.491	77.7
Gaussian regression	1.935	2.035	94.0	93.5	0.000	74.6

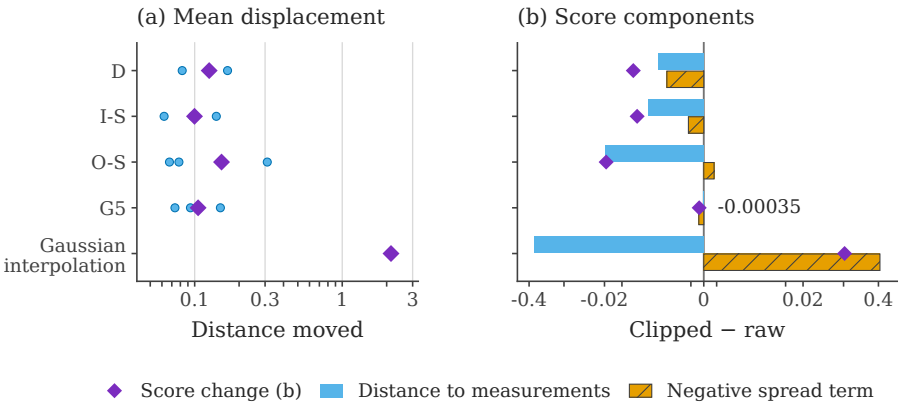


Figure 1: Effect of clipping the same draws. Diffusion predictions use checkpoints that minimize raw validation score. (a) Mean distance moved in 60 standardized coordinates, with a logarithmic axis. Blue circles denote individual fits; purple diamonds show their mean or the result for one Gaussian ensemble realization. (b) Solid bars give the change in distance to withheld values; striped bars give the change in the negative of half the mean distance between draws. Purple diamonds give their sum, the change in energy score. The symmetric logarithmic axis is linear inside ± 0.005. Clipping changes D's energy score by -0.0064; choosing a separate checkpoint adds -0.0118.

Clipping moves G5 draws in the standardized primary coordinates with little change in energy score (Table 13). For Gaussian interpolation, both the mean distance to withheld values and half the mean distance between draws decrease. The larger decrease in the subtracted term makes the energy score worse. D's change in Table 1 combines clipping with selection of a different checkpoint (Figure 1).

At the final training step, the original O-S coefficient gives a lower mean raw energy score than D and improves all three seeds. Model selection thus changes the D/O-S comparison. Doubling validation DDIM evaluations from 25 to 50 preserves all six raw D/O-S checkpoint choices (Appendix C.3). Table 14 gives the sensitivity results at the same final training step.

3.2 An exact denoiser example

A stationary hidden Markov model isolates the residual estimator from denoiser fitting error (Figure 3). Its exact denoiser integrates over the unknown switching probability at every evaluation. For 36 replicate contexts at $t = 100$, two denoiser evaluations give mean absolute $R_{\text{obs}} = 4.17 \times 10^{-5}$ in HMM units and energy score 6.958. Independent draws from the exact conditional distribution give an energy score of 4.164. In a separate evaluation on 288 HMM windows, two denoiser evaluations also score worse than exact conditional draws, while DDIM with 25, 50, and 100 evaluations gives scores much closer to the score of exact draws on those windows (Table 2). These Monte Carlo comparisons have finite sampling error.

4 Conclusion

The variance identity constrains combinations of magnetic statistics, leaving their conditional distribution to be learned. The penalty's benefit depends on checkpoint choice, even with equal validation and test sampling effort. Most of G5's mean energy score improvement over D comes from greater ensemble spread. G5's coverage is below nominal. For scalar variance, most observations outside its intervals fall below them at two seeds and above them at the third. Because G5 changes both representation and loss, separating their effects requires further experiments. Distributional scores and coverage must accompany constraint checks and explicit checkpoint selection. The evidence covers hours masked in complete records, three seeds and a test period examined earlier. Naturally missing records, later years, other interval lengths and further training seeds require separate evaluation.

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A Physically possible magnetic statistics and energy

The realizability conditions follow from the possible sums of five vectors. Let $a_j = \|B_j\|$, so $\sum_j a_j = 5m$. Vectors with these lengths can sum to a vector of length $5r$ precisely when $\max(0, 2\max_j a_j - 5m) \le 5r \le 5m$. Thus each a_j must be at most $5(m+r)/2$. With the total length fixed, the sum of squared lengths is largest when two lengths are $5(m+r)/2$ and $5(m-r)/2$ and the other three are zero. Hence $m^2 + v_s \le 5(m^2+r^2)/2$, which gives the upper bound in Equation 3.

Equal lengths attain the lower bound $v_s = 0$. The nonnegative lengths have a fixed sum and a common upper bound, so their feasible set is convex and contains both extremes. Continuity of the sum of squared lengths then gives every intermediate value. Rotating a nonzero vector sum gives the mean the direction of μ; a zero sum already has the required mean. The conditions are therefore both necessary and sufficient, including when all five vectors vanish.

The quantity $Q = m^2 + v_s = \|\mu\|^2 + v_v$ is the mean squared magnitude of the five vectors, each already averaged over one minute. Multiplying by $10^{-18}/(2\mu_0)$ converts nT^2 to the magnetic energy density represented by those averages. This estimate excludes fluctuations lost in the original minute averages.

Averaging Q over a longer interval requires only the statistics of its constituent bins. The check uses all 97,699 eligible intervals of 15 minutes, each starting on a UTC quarter hour within the union of training windows. Combining the three sets of statistics for five minutes agrees with a direct calculation from the original fifteen vectors to a maximum absolute difference of $9.09 \times 10^{-13}\,\mathrm{nT}^2$. Using only the three mean vectors loses $0.8943\,\mathrm{nT}^2$ on average.

B Data sources and selection

Recomputing the magnetic statistics from the original OMNI inputs reproduces their values, normalization and division into training, validation and test sets exactly. The windows also retain their original IDs and order.

The magnetic inputs are the vector components in geocentric solar ecliptic coordinates, GSE, listed as NASA words 15–17. The modified OMNI format identifies Wind as source 52 and ACE as source 71. NASA word 14 contains the average magnitude, which can differ from the magnitude of the averaged vector components used here. Before aggregation over five minutes, data checks reject nonfinite values and fill codes from the OMNI specification. Calendar fields must be integral, timestamps unique and in order, and the spacecraft source common to the whole window. No amplitude cutoff or additional removal of spikes based on the values to be predicted is applied.

NASA estimates arrival times and forms weighted averages for each minute before the OMNI record is masked. Neighboring output minutes can share measurements, and the phase front normals used to estimate propagation depend on neighboring magnetic observations. Any contribution of the hidden interval to those earlier estimates remains after masking. NASA retains arrivals whose order changes after propagation and describes the resulting mixing of plasma regions [8, Sections 3a–3c and Appendix 5]. A common spacecraft source within each window prevents source changes, while arrival order can still differ from observation order. Directional cancellation, variance and changes at gap edges can reflect both this processing and solar wind variability.

Data completeness depends on propagation as well as solar wind conditions. Magnetic observations can be excluded when suitable plasma velocities are missing or when no acceptable phase front normal spans a long interval with little variation. Completeness alone therefore cannot reliably distinguish quiet from disturbed solar wind. NASA selects spacecraft data in segments of three hours using coverage and spatial separation, with a preference for Wind from 2020 onward [8, Appendix 6]. Comparisons between spacecraft groups consequently combine differences in instruments, dates and processing choices. The results use the original OMNI inputs; comparison with NASA's current release remains outstanding.

The training dates contain 4,816,800 recorded minutes, including 4,400,682 with valid magnetic vectors. These counts include minutes outside eligible training windows. Among the 4,298,363 valid magnetic minutes with a known duration between observing times, DBOT, 555,719 have negative DBOT, indicating an apparent reversal of observation order at the spacecraft. Of the 300,392 distinct eligible bins, 55,339 contain at least one such minute. The phase front normal interpolation field gives

the percentage of contributions to a magnetic average over one minute whose normal was interpolated [6]. This field concerns the propagation normals. These processing measures leave the set of test windows intact.

Of the 957,720 possible starts for training windows, 122,309 meet the data requirements. The denominator counts starts on the grid of five minutes whose full four hours lie inside a calendar month assigned to training. These windows contain 163,988 distinct bins that can occupy a hidden position in at least one window. Availability varies by year: 11,340 starts remain in 2005 and 2,568 in 2014. The role of solar activity in this missingness is undetermined.

Evaluation limits each month's contribution by first taking eligible windows of four hours in chronological order without overlap within that month. From N such windows, $q = \min(N, q_0)$ are retained at indices $\lfloor (j + 0.5)N/q \rfloor$ for $j = 0, \ldots, q - 1$, with $q_0 = 8$ for VAL and $q_0 = 12$ for TEST. The quota limits the influence of months with more complete records, although completeness still determines which windows are eligible.

C Model training, sample generation, and scoring

The main solar wind comparison has twelve diffusion fits, each trained for 29,636 optimizer updates with batches of 128. Each update uses the loss gradient from one batch. Training makes 31 passes through the data, including the shorter batch at the end of each pass.

The U-Net has widths 64/128/128 and contains 1,603,398 parameters when it generates six coordinates, or 1,603,013 for G5. The architecture omits attention and dropout. Evaluation uses the fitted weights directly, without exponential averaging. AdamW uses learning rate 2×10^{-4}, betas $(0.9, 0.999)$, epsilon 10^{-8}, weight decay 10^{-4}, and a limit of 1 on the norm of the total gradient.

The model state is saved at 2,868, 9,560, 16,252, 22,944, and 29,636 optimizer updates. These five checkpoints allow comparisons at common stages of training.

The cosine diffusion schedule has offset 0.008 and an upper limit of 0.999 for beta. Its posterior variance is $\Sigma_t = \beta_t(1 - \bar{\alpha}_{t-1})/(1 - \bar{\alpha}_t)$, with $\Sigma_1 := \Sigma_2$. Observations and score calculations use float64; network calculations and diffusion samples use float32. For evaluation, PCG64 supplies initial noise for six coordinates, of which G5 uses the first five. Training uses a CPU PyTorch generator for timesteps and Gaussian noise in the generated coordinates. The same available measurements are supplied at every denoiser evaluation.

The energy score is also calculated for the vector components alone and for all six coordinates. Other measures include the continuous ranked probability score (CRPS), interval coverage and width, identity residuals, failures of the physical inequalities, and errors at gap edges relative to the withheld observations.

G5 predicts five coordinates, so its data loss averages squared error over 60 hidden coordinates; the other diffusion models use 72. Removing the v_v prediction changes both the weight of each remaining coordinate in that average and the network input and output dimensions. This comparison leaves their individual contributions to the score difference unresolved. Algebraic completion alone preserves the primary score when the first five coordinates stay fixed.

Prediction intervals are formed separately for each coordinate from its 39 sorted draws, including both endpoints. Counting from one, draws 2/38 define the 90% interval and draws 10/30 define the 50% interval. If the withheld value and ensemble members are independent draws from the same continuous conditional distribution, rank symmetry gives expected coverage $(h - l)/40$ for endpoints l and h. Ties in the discrete HMM distribution can raise the coverage of these inclusive intervals.

Errors at gap edges compare the predicted changes between hidden and available bins with the actual changes, separately from changes inside the gap. For interpolation, the mean-vector changes at both edges are $(\mu_{30} - \mu_{17})/13$, which can be nonzero. The Gaussian model also permits nonzero residuals at those edges. The identity residual measures departure from the equality; the fractions in the tables count failures of the physical inequalities.

The mean estimator M evaluates the residual at the first denoiser prediction of the data before noise was added. S makes a DDIM transition from t to 1 without new random noise, then predicts the data again at that lower noise level. Both estimators reverse the affine standardization to return the

prediction to physical units before evaluating the residual. At $t = 1$, S reuses the first prediction, so M and S coincide. S provides an approximation to a conditional draw.

Gradients of the two loss terms are evaluated on the same batches. For each O-S seed, the calculation uses the same eight TRAIN windows and eight VAL windows at the first and final common checkpoints, with noise seed 7003 and $c_R = 0.0005$. Both residuals and both estimators are evaluated at $t \in \{1, 25, 50, 100\}$ while the model weights remain constant. Norms and angles refer to gradients of the weighted data and residual terms before clipping of the total gradient. Undefined ratios and angles remain missing. These measurements characterize the loss gradients on those batches; the cause of differences in test scores remains undetermined.

C.1 How the residual penalty is weighted

The relative weight of the residual penalty varies with the diffusion timestep, even at $c_R = 0.0005$. With $\lambda_t = c_R/(2\Sigma_t)$, the ratios λ_t/w_t at $t = 1, 25, 50, 100$ are 0.1239, 0.004183, 0.008642, and 1030.7. At the final noise level, the Min-SNR data weight is only 2.429×10^{-7}. These ratios compare loss coefficients; the corresponding gradient ratios also depend on the derivatives of the two terms.

At the final O-S checkpoints, the norm of the residual gradient divided by the norm of the data gradient ranges from 315 to 1799 on TRAIN and 214 to 1266 on VAL at $t = 100$. At $t = 25$, the ranges are 0.0081 to 0.0267 and 0.0103 to 0.0465. Each range spans three seeds, with one calculation on eight windows per seed. The absolute TRAIN residual-gradient norms at $t = 100$ are 7.379×10^{-5} to 4.401×10^{-4}, much smaller than the 0.1396 to 0.4587 range at $t = 1$. Thus the large ratio at high noise accompanies a small absolute gradient. These measurements precede clipping of the total gradient, and optimizer updates combine noise levels. The residual term's contribution to prediction quality remains unresolved.

C.2 Choosing checkpoints and prior use of test data

The solar wind comparison uses new fits because the earlier training records do not identify the code that produced those models. The current implementation also corrects two evaluation errors: the earlier sampler supplied different values outside the gap from those used during training, and the earlier HMM calculation received a switching probability unavailable to the prediction methods. The changed fits prevent a separate estimate of either repair's effect on the earlier scores. The test period had already been examined before this comparison.

D at seed 7 is the only fit for which clipping changes the checkpoint chosen by validation: raw predictions choose 9,560 updates, while clipped predictions choose 29,636. For D, the mean energy score across the three training seeds is 1.881 when clipping uses the checkpoints chosen from raw predictions. Choosing each checkpoint by its clipped score gives a mean of 1.869.

The original rule for superiority compares raw O-S with four procedures: raw D, raw G5, and D and G5 whose checkpoints are chosen independently for clipping. O-S must have a mean energy score at least two percent lower than each procedure, both difference intervals below zero, and the same direction of difference at all three seeds. A separate rule asks whether the raw O-S versus raw D score difference lies within a prescribed range: both relative intervals must be strictly inside $[-0.02, 0.02]$, with positive D score denominators in every resample. Both rules concern score differences; equality of the predictive distributions remains undetermined.

For O-S minus D, resampling groups of two consecutive years gives an interval of $[-0.0345, 0.0233]$. The intervals describe temporal variation separately for each comparison, conditional on the existing fits and draws. They omit adjustment for multiple comparisons and variation from new training seeds.

C.3 Scores during training and the number of sampling steps

The later checkpoint analysis tests whether doubling the number of denoiser evaluations during validation changes model choice. It covers all five checkpoints for raw D and O-S at each of the three training seeds, using the original weights, windows, 39 draws and evaluation seed 7. The original validation ensembles use 25 denoiser evaluations; 30 new ensembles use 50 with the same initial noise. TEST then covers the same 30 model states at 50 evaluations, reusing ten ensembles after checking their identities and generating twenty more. Figure 2 includes all five checkpoints.

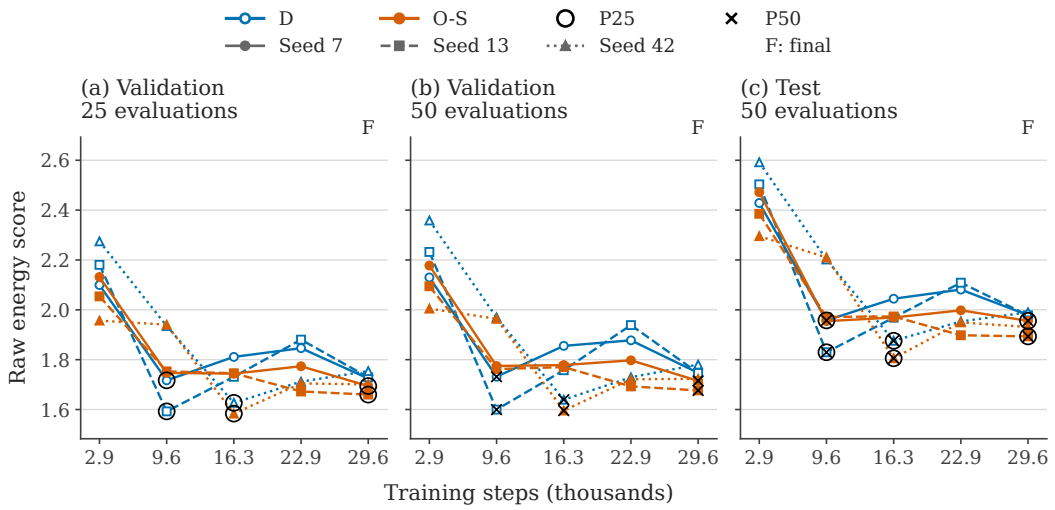


Figure 2: Scores across all five checkpoints for D and O-S at the three training seeds. The subplots compare original VAL ensembles with 25 DDIM evaluations, new VAL ensembles with 50, and TEST ensembles with 50. Each point averages windows within months and then gives months equal weight. Colors distinguish methods, with open markers for D and filled markers for O-S; line styles and marker shapes distinguish seeds. Large black rings mark P25, the original validation choice, and crosses mark P50, the choice with 50 evaluations. These choices coincide for all six fits. F marks the common final optimizer update. Each ensemble has 39 draws. VAL and TEST scores concern different sets of windows. The additional TEST analysis evaluates choices already made on VAL.

Validation with 50 evaluations determines the checkpoint choices, and both the choices and evaluation count are fixed before any additional TEST samples are generated. The test period had already been examined in the earlier study.

All six validation choices agree between 25 and 50 evaluations. For these fits and this realization of the initial noise, doubling the validation evaluations leaves the comparison at the chosen checkpoints intact. At the five common update counts, mean TEST differences for O-S minus D are -0.1238, 0.0490, -0.0470, -0.0993, and -0.0554, respectively. O-S has lower scores at all three seeds at the last two counts; seed differences have mixed signs at the first three. Its advantage at the final update is part of this variation during training.

D chooses an earlier checkpoint for every seed; O-S chooses the final one for seeds 7 and 13. Replacing these choices with the final checkpoints raises D's mean TEST score by 0.0945 and O-S's by 0.0419, changing their difference from -0.0028 to -0.0554. D's larger deterioration accounts arithmetically for O-S's larger advantage at the final update. This calculation leaves the roles of regularization and overfitting unresolved.

The years represented in validation can affect which checkpoint is chosen. In 10,000 resamples of the twenty years, D at seed 7 chooses update 9,560 in 61.0% of resamples with 25 evaluations and 71.5% with 50; all other resamples choose the final checkpoint. D at seeds 13 and 42 retains its respective earlier choice in every resample under either evaluation count. Each occurrence of a sampled year contributes both validation months, and every checkpoint uses the same resampled years. The model weights and draws remain constant, so these frequencies measure variation across validation years. Variation under fresh sampling noise and performance on a new population remain undetermined. The numerical results include ranks, score margins, resampling frequencies and choices after omitting each year in turn, for both methods and both evaluation counts.

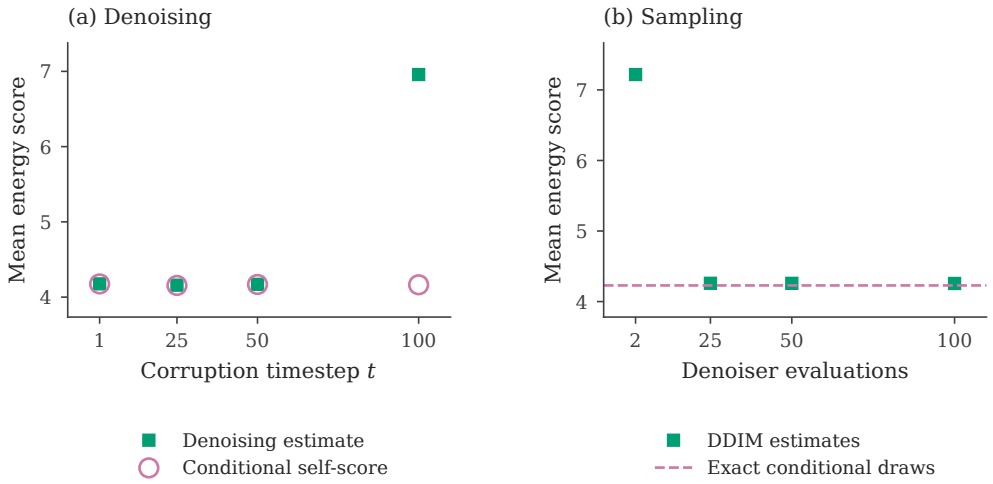


Figure 3: Denoising and sampling in the stationary hidden Markov example. (a) Every timestep uses the same 36 cases, each with specified endpoint states and vector geometry. Each case uses 128 conditional paths with independent noise; exact conditional draws use a separate random stream. Squares give mean energy scores for the denoising estimates, and open circles estimate the conditional distribution's expected score from independent draws. (b) Squares compare the procedure with 2 evaluations and DDIM using 25, 50 and 100 evaluations, each with 39 draws on the same 288 windows. The dashed line gives the score of an independent exact conditional ensemble. The two subplots use different evaluation populations.

D A stationary hidden Markov model with a known distribution

For $\alpha \in \{0, 1, 2\}$, let $c_j = j - 3$ for $j = 1, \ldots, 5$ and $d_\alpha = \sqrt{1 + 2\alpha^2}$. The two sets of magnetic statistics, z_0 and z_1, follow from

$$B_j^{(0)} = \frac{(1, \alpha c_j, 0)}{d_\alpha}, \qquad B_j^{(1)} = \frac{2(1 + 2\alpha c_j,\ 2 - 2\alpha c_j,\ 2 + \alpha c_j)}{3 d_\alpha}. \tag{6}$$

Each state has the same mean squared magnitude across geometries: $Q_0 = 1$ and $Q_1 = 4$ in HMM units for every α. Changing α alters the vector geometry and magnitude variance inside each bin while preserving these two energies and their difference. Standardization gives equal weight to all six states, with $B^2_{\text{ref}} = 2.5$, and is determined before evaluation.

The latent sequence is a two-state chain with switching probability $p \in \{0.1, 0.25, 0.5\}$ and transition matrix $T_p = \left(\begin{smallmatrix} 1-p & p \\ p & 1-p \end{smallmatrix}\right)$. Its endpoint states follow the stationary law $\Pr(a) = 1/2$ and $\Pr(b \mid a, p) = (T_p^{13})_{ab}$, and the twelve hidden states are drawn conditional on those endpoints.

In the record of four hours and 48 bins, the 18 bins before the gap repeat the left state's statistics and the 18 bins after it repeat the right state's statistics. These available intervals encode α and the endpoint states, leaving p unknown to the network. A Markov trajectory connects the endpoints through the twelve hidden states.

The exact conditional calculation marginalizes over the three switching probabilities, assigned equal prior weights. Before noisy evidence is included, the weights conditional on the endpoints are proportional to $\Pr(p)(T_p^{13})_{ab}$ for the stationary construction and to $\Pr(p)$ for the earlier design with independent endpoint states. At every denoiser evaluation, the current noisy input further updates these weights.

Writing $\kappa_\alpha = [1 + 2\sqrt{1 + \alpha^2} + 2\sqrt{1 + 4\alpha^2}]/5$, the residual for posterior probability q of state 1 is:

$$R_{\text{obs}}\big((1 - q)z_0 + q z_1\big) = q(1 - q)\frac{11/3 - \kappa_\alpha^2}{1 + 2\alpha^2}. \tag{7}$$

The coefficient is positive for $\alpha = 0, 1$ and negative for $\alpha = 2$. Thus a nondegenerate mixture has a nonzero residual in all three geometries, even though each individual state obeys the identity. Squaring the residual assigns a positive penalty to the exact posterior mean whenever both states have positive posterior probability. This magnetic example illustrates the effect of taking expectations through a nonlinear residual discussed by Bastek et al. [3].

The learned HMM comparison uses 4,096 training windows generated at seed 106 by cycling through the nine (α, p) combinations. The U-Net has widths 32/64/64, batches contain 128 windows, and each fit receives 5,000 optimizer updates at seed 7. Other optimizer and diffusion settings follow Appendix C. Evaluation uses 32 paths per switching probability, reusing each path for the three values of α. The resulting 288 windows thus share hidden state sequences across vector geometries.

Table 2 compares methods on the same 288 windows with stationary endpoint states, the same available information and 39 draws per method. The approximate samplers share their initial noise. The first row compares independent draws from the exact conditional distribution with the hidden values. It estimates the minimum expected score under that law, subject to Monte Carlo error. A finite estimate can lie above or below this expectation. Earlier windows with independent endpoint states are part of a separate sensitivity analysis and are excluded here.

Table 2: Mean primary energy scores on the same 288 hidden Markov windows with stationary endpoint states. The exact conditional calculation averages over the unobserved switching probability. Procedures using an exact denoiser follow DDIM. Each score is an estimate from finite ensembles in this evaluation.

Procedure	Energy score
Exact conditional draws	4.2287
Two evaluations of exact denoiser	7.2183
Exact denoiser, 25 evaluations	4.2591
Exact denoiser, 50 evaluations	4.2586
Exact denoiser, 100 evaluations	4.2567

One calculation conditions on a particular noisy input. A second measures variation over new paths with the same available states and vector geometry: each repetition draws a new conditional hidden path before adding noise, and exact conditional draws use an independent random stream. Each calculation uses 36 cases per timestep with specified available states and geometry, or 144 across four timesteps. At low noise, a posterior concentrated on one path can be correct for that particular input.

D.1 Training diffusion models on the same HMM

Table 3 compares nine diffusion fits to the stationary chain, with different residual objectives and 50 DDIM evaluations for sampling. Earlier models with independent endpoint states are excluded. M and S denote the mean estimator and the estimator using two evaluations from Appendix C. At both coefficients, models with the identity penalty have primary scores close to D, while penalties that omit the variances give higher scores with both estimators. The small differences among models with the identity penalty concern one training seed and its existing draws. Variation across new training seeds remains unquantified.

The residual estimate used in the training penalty and the final sampler assess different parts of the procedure. With 50 sampling evaluations, O-S avoids the poor score obtained from only two evaluations of the exact denoiser. The calculation with two evaluations tests the approximation used during training; the full sampler tests the resulting predictive distribution. A small residual can therefore be a poor indicator of prediction quality even when training with that penalty produces samples with good scores.

For exact conditional draws, inclusive intervals cover 100.0% and 97.49% of the primary coordinates under the nominal 90% and 50% endpoint conventions. Their mean widths are 1.4581 and 1.3485 in standardized coordinates. Ties in this discrete distribution can raise the coverage of inclusive intervals. Coverage expected for a continuous distribution is therefore insufficient to assess calibration here. The interval calculations use the original random stream and reproduce all 288 original energy scores before computing coverage and width.

Table 3: Diffusion models for the hidden Markov example after 5,000 optimizer updates at seed 7, evaluated on 288 windows with 39 draws each. Clipping uses the same model state and draws. Raw $|R|$ is the mean absolute residual of the observation identity in HMM units. The inequality column is the percentage of generated bins failing a physical inequality beyond rounding tolerance; it excludes the equality test. Coverage averages the five primary coordinates and twelve positions. The separate exact conditional ensemble has 100% coverage under the inclusive nominal 90% interval convention.

		Energy score		Raw samples		
Procedure	c_R	Raw	Clipped	$\lvert R\rvert$	Bins failing inequalities (%)	90% coverage (%)
D	0	4.2607	4.2609	0.0126	27.5	94.5
I-M	0.0005	4.3283	4.3285	0.2861	26.6	84.3
I-M	0.005	4.6021	4.6022	0.7050	26.6	81.2
I-S	0.0005	4.4991	4.4993	0.6262	26.0	77.5
I-S	0.005	4.8020	4.8020	1.0086	29.2	74.6
O-M	0.0005	4.2604	4.2605	0.0122	27.8	94.7
O-M	0.005	4.2544	4.2545	0.0095	27.4	95.3
O-S	0.0005	4.2593	4.2595	0.0120	27.8	94.8
O-S	0.005	4.2551	4.2553	0.0094	27.4	95.9

E Variation across seeds, interval coverage, and alternative scores

In this appendix and Appendix F, Gaussian denotes Gaussian interpolation: a residual drawn from a Gaussian fit on TRAIN is added to interpolation between the adjacent observations. Gaussian regression uses the wider available context and is reported separately in Table 1 and Appendix G.

The primary energy score measures predictive error over the whole missing hour. Each of the five primary quantities is centered and divided by its TRAIN scale before the twelve positions are combined into 60 coordinates. Standardization puts the vector components and mean magnitude, measured in nT, and scalar variance, measured in nT^2, on a common dimensionless scale. The score subtracts half the mean distance between distinct ensemble members from their mean distance to the withheld measurements. Both terms must be considered together: a larger spread term alone cannot establish better predictions. The energy score is a statistical loss. Magnetic energy density is a distinct physical quantity.

Each seed in Table 4 identifies one training run. Every fit reaches the same final update, but validation can select an earlier checkpoint. Raw and clipped procedures select checkpoints independently within that run. Coverage is calculated separately for each ensemble, without pooling draws across seeds.

The intervals in Table 5 instead measure variation across the test period while holding the fits and their draws fixed. At seed 7003, 10,000 resamples draw with replacement from either the twenty years or the ten consecutive groups of two years, 2005–2006 through 2023–2024. Both methods in a comparison use the same sampled years. Each occurrence of a repeated year contributes its months again, preserving the intended month weights.

For O-S minus D, the relative intervals are $[-0.016766, 0.014049]$ and $[-0.016842, 0.012995]$, respectively. Both lie within the original $\pm 2\%$ comparison range for these procedures. The O-S versus I-S and I-S versus D intervals allow both improvement and deterioration. These marginal intervals describe the existing models and draws and use only twenty years or ten groups. They exclude variation from new fits. Stationarity across solar cycles and nominal interval coverage over repeated studies are not established.

E.1 Interval coverage, width, and score contributions

In Table 6, O-S's absolute departure from nominal 90% coverage exceeds D's by 2.8 percentage points, beyond the prespecified two-point threshold. Gaussian interpolation has coverage near nominal for its 90% intervals but substantial overcoverage for its 50% intervals. Calibration conditional on the available measurements requires further assessment.

Table 4: Training updates chosen by validation, primary energy score in standardized coordinates, and raw nominal 90% coverage for each seed. Raw and clipped procedures select checkpoints independently. Coverage is calculated within each ensemble of 39 draws and averages the five primary channels and twelve hidden positions.

		Chosen updates		Energy score		Raw coverage
Method	Seed	Raw	Clipped	Raw	Clipped	(%)
D	7	9560	29636	1.9575	1.9147	67.2
D	13	9560	9560	1.8297	1.8213	75.9
D	42	16252	16252	1.8758	1.8724	75.4
I-S	7	9560	9560	1.9423	1.9385	63.3
I-S	13	22944	22944	1.8954	1.8826	66.5
I-S	42	16252	16252	1.8070	1.8071	78.2
O-S	7	29636	29636	1.9558	1.9116	60.4
O-S	13	29636	29636	1.8934	1.8824	70.5
O-S	42	16252	16252	1.8056	1.8045	79.2
G5	7	29636	29636	1.7236	1.7239	75.9
G5	13	29636	29636	1.7853	1.7889	67.1
G5	42	16252	16252	1.8453	1.8404	84.0

Table 5: Differences in primary energy score, calculated as the first method minus the second. Procedures without a superscript use raw draws; c denotes clipping with a checkpoint selected independently on validation data. The first four rows assess O-S superiority; the remaining comparisons are secondary. Year and Two-year report marginal percentile 95% intervals from paired resampling of years or consecutive groups of two years. Seed differences are ordered 7, 13, 42. Gaussian denotes Gaussian interpolation.

Contrast	Mean	Year	Two-year	Seed differences
O-S − D	-0.0028	$[-0.0346, 0.0242]$	$[-0.0345, 0.0233]$	$(-0.0018, 0.0637, -0.0702)$
O-S − D^c	0.0154	$[-0.0024, 0.0312]$	$[-0.0034, 0.0320]$	$(0.0410, 0.0721, -0.0669)$
O-S − G5	0.1002	$[0.0712, 0.1391]$	$[0.0671, 0.1422]$	$(0.2322, 0.1080, -0.0398)$
O-S − G5^c	0.1005	$[0.0716, 0.1396]$	$[0.0677, 0.1420]$	$(0.2319, 0.1045, -0.0348)$
O-S − I-S	0.0033	$[-0.0236, 0.0349]$	$[-0.0259, 0.0431]$	$(0.0135, -0.0020, -0.0015)$
I-S − D	-0.0061	$[-0.0562, 0.0330]$	$[-0.0720, 0.0410]$	$(-0.0152, 0.0656, -0.0688)$
O-S − Gaussian	-0.2402	$[-0.5298, -0.0481]$	$[-0.4539, -0.0882]$	$(-0.1694, -0.2318, -0.3196)$

Table 7 gives coverage by channel: O-S covers scalar variance more often than D, but covers each vector component and mean magnitude less often. Similar primary energy scores can conceal these differences because the score evaluates a distribution over all 60 coordinates, whereas coverage checks each coordinate and position separately.

At the checkpoints chosen using raw validation draws, G5 has a slightly smaller mean distance to the withheld measurements than D and a larger spread term. The score subtracts the spread term, whose increase accounts for most of the mean improvement in Figure 4(a). Both terms use the same 60 standardized coordinates and equal month and seed weights.

The two score terms change differently across seeds. At seed 13, G5 decreases both distance to the measurements and spread, with the larger decrease in distance. At seed 42, both increase, with the larger increase in spread. Both patterns lower the score. This decomposition explains the score differences for the existing predictions. It cannot identify a training mechanism or separate G5's changes to representation and loss.

Where observations fall outside the intervals. G5's nominal 90% intervals still miss many observations, especially for v_s, whose coverage is 40.5%. To distinguish the direction of these errors, the tail calculation counts observations strictly below the second sorted draw, inside the inclusive interval, and strictly above the 38th. If the observation and all 39 ensemble members are independent draws from the same continuous conditional distribution, their exchangeable ranks imply expected fractions of 5%, 90%, and 5%. A fitted model's coverage must be measured.

Every TEST window contributes to the calculation. For each fit, the calculation averages positions within a window and windows within a month, then gives months equal weight. Results are averaged equally across fits of the same method. Coverage checks for individual variables complement the proper score [14] and can expose errors that an overall coverage percentage can hide.

Averaged over the three fits, 49.0% of observations lie below G5's v_s intervals and 10.5% above them. The seed dots in Figure 4 show a reversal that the averages obscure: errors lie predominantly below the intervals at seeds 7 and 13, and above them at seed 42. D also reverses direction at seed 42. O-S's mean fractions are 37.7% below and 8.4% above. Wider intervals still leave many observations uncovered.

For G5's v_s intervals, the fraction below exceeds the fraction above by 38.4 percentage points, with a 95% interval of $[36.2, 40.6]$ from resampling years. The difference is calculated before rounding the two tail percentages. Each of 10,000 resamples calculates both fractions on the same sampled years and retains every occurrence of a repeated year. The interval concerns this one contrast, conditional on the existing models, checkpoints and draws; it excludes variation from new training seeds or a new checkpoint choice.

Gaussian regression covers 98.1% of v_s observations, with 0.6% below and 1.2% above its intervals. Its wide intervals accompany a worse primary energy score. Coverage, interval width and score describe different properties of the predictive distribution and should be read together. Both temporal resampling schemes include every method and coordinate.

Table 6: Marginal continuous ranked probability scores and predictive intervals for the five standardized primary coordinates. Scores and widths use dimensionless standardized values; coverage is in percent. Results average hidden positions, months and the three diffusion seeds. Gaussian interpolation, interpolation and persistence each use one existing ensemble. Clipped rows apply the clipping map. Raw and clipped diffusion procedures select validation checkpoints independently, except the D row that retains the checkpoint selected using raw draws. Gaussian denotes Gaussian interpolation.

Procedure	Continuous ranked probability score	50% intervals		90% intervals	
		Coverage (%)	Width	Coverage (%)	Width
D raw	0.1753	35.5	0.2878	72.8	0.8709
D clipped	0.1721	33.5	0.2814	71.0	0.8617
D clipped, raw weights	0.1734	36.1	0.2883	73.5	0.8722
I-S raw	0.1745	32.7	0.2785	69.4	0.8315
I-S clipped	0.1735	32.8	0.2789	69.6	0.8319
O-S raw	0.1729	31.9	0.2691	70.0	0.8213
O-S clipped	0.1701	32.6	0.2685	71.1	0.8195
G5 raw	0.1634	38.1	0.3137	75.7	0.9308
G5 clipped	0.1624	38.4	0.3128	75.8	0.9287
Gaussian raw	0.1960	76.4	0.6262	94.3	1.5822
Gaussian clipped	0.1935	68.6	0.5512	93.6	1.3933
Interpolation raw	0.1993	0.0	0.0000	0.0	0.0000
Interpolation clipped	0.1993	0.0	0.0000	0.0	0.0000
Persistence raw	0.2697	0.1	0.0000	0.1	0.0000
Persistence clipped	0.2697	0.1	0.0000	0.1	0.0000

The vector energy score in Table 8 uses 36 standardized coordinates; the six-coordinate score uses 72. Adding the redundant v_v changes distances between outcomes and can change method rankings, even though v_v is determined by the other quantities for realizable statistics. The absolute values of these two scores reflect different distance geometries and represent different losses. Gaussian interpolation has a lower mean vector score than D, I-S and O-S. G5 has the lowest mean five-coordinate primary score and changes rank when v_v is included.

For each draw, resolved magnetic energy density is calculated from $m^2 + v_s$ by converting nT^2 to T^2 and dividing by $2\mu_0$. Averaging over the hidden hour gives the scalar prediction that is then scored. This physical quantity excludes fluctuations within each vector average over one minute and is distinct from the statistical energy score. Its continuous ranked probability score (CRPS) and interval width

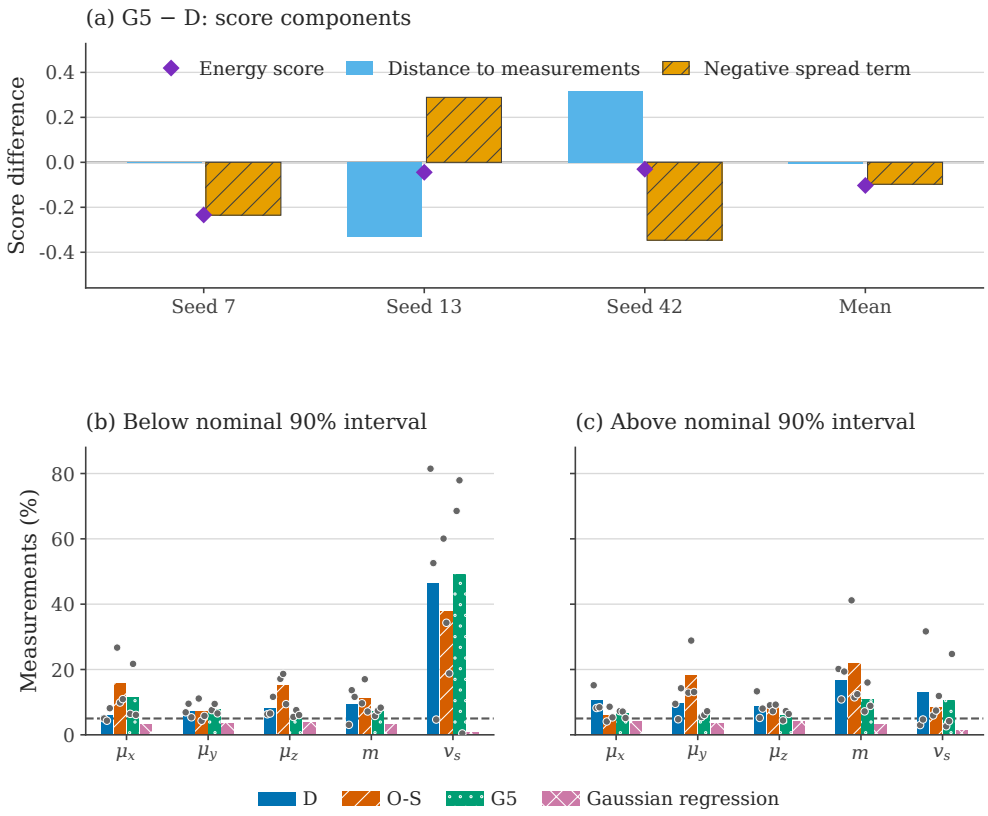


Figure 4: Score contributions and interval errors for the original ensembles, using diffusion checkpoints selected by raw validation score. (a) G5 minus D for each seed and their mean. The change in distance to withheld measurements minus the change in the spread term equals the difference in energy score. Both terms use 60 coordinates standardized on TRAIN; the spread term is half the mean Euclidean distance between distinct draws. (b,c) Fractions of observations below and above nominal 90% intervals for the five primary coordinates. Bars average three diffusion fits or show one Gaussian realization. Gray dots within each diffusion bar give seeds 7, 13 and 42 from left to right. The dashed 5% reference assumes continuous exchangeability of the observation and ensemble members. Months receive equal weight. These comparisons use the existing TEST ensembles and were made after the original results were available.

have units of J m^{-3}, shown in the table as multiples of 10^{-12}. Gaussian interpolation has the lowest CRPS for this quantity among the procedures in Table 8. Clipping enforces the moment inequalities to rounding tolerance. Its effects on these distributional scores can have either sign.

E.2 Vector increments at the edges and inside the missing interval

The increments at each gap edge measure the change between available and generated vectors: indices 17 to 18 at the left edge and 29 to 30 at the right. Table 9 reports their mean magnitudes and the error of individual draws. RMSE is calculated across the 39 draws within each window, then averaged over months and seeds. Because this RMSE depends on both predictive spread and bias, a smaller value alone does not establish a better conditional distribution. The error of the ensemble mean is a different quantity.

Table 7: Raw nominal 90% coverage (%) for each primary channel. Hidden positions, months and diffusion seeds receive equal weight in their respective averages. Gaussian denotes Gaussian interpolation.

Procedure	μ_x	μ_y	μ_z	m	v_s
D	83.6	83.2	83.0	73.8	40.6
I-S	82.2	74.8	80.7	74.3	34.9
O-S	78.2	74.6	76.4	67.0	53.9
G5	82.1	85.9	87.6	82.2	40.5
Gaussian	93.1	93.5	92.2	94.1	98.5
Interpolation	0.0	0.0	0.0	0.0	0.0
Persistence	0.1	0.2	0.1	0.0	0.0

Table 8: Alternative scores for magnetic coordinates and resolved magnetic energy density. Vector and six-coordinate energy scores use 36 and 72 standardized coordinates, respectively. For mean energy density over the missing hour, continuous ranked probability score (CRPS) and interval width are in 10^{-12} J m^{-3} and coverage is in percent. The density uses vector averages over one minute and excludes energy from variation within each minute. Raw and clipped diffusion procedures select validation checkpoints independently. Gaussian denotes Gaussian interpolation.

	Energy score		Resolved magnetic energy density		
Procedure	Vector	Six coordinates	Continuous ranked probability score	90% coverage (%)	90% width
D raw	1.6351	2.1528	2.8599	74.3	6.2166
D clipped	1.6222	2.5726	2.7568	68.5	6.6024
I-S raw	1.6435	2.1590	2.4664	72.4	6.6657
I-S clipped	1.6435	2.8791	2.3049	73.1	6.8576
O-S raw	1.6316	2.1503	2.9225	65.3	6.5977
O-S clipped	1.6316	2.3863	2.7461	72.2	6.6863
G5 raw	1.5740	2.4959	1.8961	82.2	6.8060
G5 clipped	1.5740	2.4285	1.9066	82.4	6.8757
Gaussian raw	1.5909	2.7342	1.0816	93.8	6.1012
Gaussian clipped	1.5909	4.2146	2.5465	64.8	11.2941
Interpolation raw	1.9834	2.8920	1.2771	0.0	0.0000
Interpolation clipped	1.9834	4.6381	1.2771	0.0	0.0000
Persistence raw	2.7925	3.6601	2.6284	0.0	0.0000
Persistence clipped	2.7925	3.6601	2.6284	0.0	0.0000

The withheld record's mean increment magnitudes are 1.0460 nT at the left edge, 1.0539 nT at the right edge, and 1.0831 nT inside the missing hour. Interpolation produces smooth increments by construction. Recovery of missing transients must be assessed against the withheld measurements; smoothness alone is insufficient.

The variogram score checks differences between coordinates, with exponent $1/2$ [15]. Here it averages equally over 175 pairs of standardized primary coordinates: 55 pairs at adjacent times within a channel and 120 pairs between channels at the same time. For pair (i, j), define $d_k = |x_{k,i} - x_{k,j}|^{1/2}$ for ensemble member k. Its contribution is $(|y_i - y_j|^{1/2} - \bar{d})^2 - s_d^2/K$, where $K = 39$ and s_d^2 uses divisor $K - 1$. The squared term compares the observed coordinate difference with its ensemble estimate. For independent draws, the variance of that estimate introduces upward Monte Carlo bias; subtracting s_d^2/K removes this bias in expectation. The final score is the mean contribution over the 175 pairs.

E.3 Numerical precision when transforming observations

Table 10 measures arithmetic error in the 5,760 hidden observation bins, without evaluating a model. The affine transform is fitted on TRAIN and applied in float64. One route retains float64 values; the other stores them in float32 before the scoring decoder performs the inverse transform in float64. The

Table 9: Vector increments and variogram scores from raw draws. Mean increment magnitudes and RMSE across draws within each window are in nT. The variogram uses the 175 pairs of standardized primary coordinates defined in the text, with the correction for bias from a finite ensemble. At fixed model weights, clipping preserves vector increments at both edges and inside the missing hour. Changes to m and v_s can still alter the variogram. D's clipped procedure also changes its vector predictions because validation selects a different checkpoint for seed 7. Gaussian denotes Gaussian interpolation.

		Mean increment magnitude			Root mean square error	
Procedure	Variogram	Left edge	Right edge	Internal	Left edge	Right edge
D	0.0524	1.3121	1.3581	0.8560	1.8164	1.8323
I-S	0.0521	1.3309	1.4224	0.8156	1.8059	1.8854
O-S	0.0524	1.3111	1.3296	0.7635	1.7974	1.7990
G5	0.0490	1.3717	1.3830	1.0789	1.8893	1.8958
Gaussian	0.0905	1.5737	1.5587	1.5718	2.1020	2.1114
Interpolation	0.0658	0.2518	0.2518	0.2518	1.0373	1.0309
Persistence	0.1003	0.0000	3.2730	0.0000	1.0460	3.1140

Table 10: Numerical error after normalization and its inverse. Maximum $|\Delta u|$ is the largest absolute change among the five standardized primary coordinates in all hidden bins. Equality residuals are in nT^2. Deriving v_v makes the computed residual zero without removing rounding error from the primary coordinates.

Representation	Maximum $\lvert\Delta u\rvert$	Mean $\lvert R_{\text{obs}}\rvert$	Maximum $\lvert R_{\text{obs}}\rvert$
Original float64	0	8.476×10^{-15}	3.326×10^{-13}
Float64 normalization and inverse	1.271×10^{-15}	8.987×10^{-15}	5.573×10^{-13}
Float32, six coordinates	4.977×10^{-7}	1.571×10^{-6}	6.268×10^{-5}
Float32, derived v_v	4.977×10^{-7}	0	0

route using five coordinates then derives v_v. All routes satisfy the inequalities for these observations, even without a rounding allowance. Exact algebraic completion can make the computed equality residual zero while leaving rounding error in the primary coordinates.

F Variation across spacecraft, years, and magnetic conditions

Within each TEST subgroup in Table 11, the primary scores give equal weight to the months represented. Because the represented months differ among groups, weighting subgroup means by their window counts need not recover the overall test score.

The spacecraft comparison also reflects calendar differences: in 2020–2024, 119 windows use Wind and only one uses ACE. These subsets cannot separate spacecraft from calendar period, and the calendar groups combine different phases of the solar cycle. Intervals from resampling years remain exploratory, conditional on the existing predictions and unadjusted for the number of comparisons.

The magnitude and increment groups use information from the missing hour. The 480 TEST windows are divided into thirds by mean withheld magnitude, with thresholds 3.9350 and 5.7891 nT, or by mean internal vector increment, with thresholds 0.6134 and 1.2055 nT. Each group contains 160 windows and there are no threshold ties.

Selecting cases by their withheld outcome changes the distribution of outcomes being scored without changing the forecasts issued for those cases. Even a correct conditional predictive distribution can then lose a comparison using a proper score within a selected group. The opposite signs of O-S minus D at low and high magnitude describe these particular cases; a prediction benefit conditional on field strength requires an assessment that accounts for the selection. Each annual result in Table 12 covers one year; intervals from resampling years require variation across years.

Table 11: Raw primary energy score by spacecraft, calendar period, and TEST thirds defined by withheld mean magnitude or mean internal vector increment. The last column gives a marginal 95% interval for O-S minus D from resampling years. Groups defined by withheld outcomes describe the selected cases. Establishing superiority of a conditional predictive distribution requires accounting for that selection. Scores use standardized primary coordinates and average the same three diffusion fits. ACE denotes the Advanced Composition Explorer.

Stratum	Windows	D	O-S	G5	O-S − D	Year interval
Wind (52)	335	1.7136	1.7119	1.6159	-0.0017	$[-0.0382, 0.0303]$
ACE (71)	145	1.8597	1.9008	1.7886	0.0411	$[-0.0009, 0.0869]$
2005–2009	120	1.9817	2.0219	1.9051	0.0403	$[0.0158, 0.0638]$
2010–2015	144	1.7735	1.7728	1.6890	-0.0007	$[-0.0416, 0.0415]$
2016–2019	96	1.6349	1.6220	1.5657	-0.0129	$[-0.0543, 0.0237]$
2020–2024	120	2.1330	2.0928	1.9545	-0.0403	$[-0.1384, 0.0293]$
Magnitude, lower third	160	1.1777	1.2324	1.1010	0.0547	$[0.0195, 0.0835]$
Magnitude, middle third	160	1.4606	1.4747	1.4339	0.0140	$[-0.0044, 0.0310]$
Magnitude, upper third	160	3.3910	3.3366	3.1849	-0.0544	$[-0.1089, -0.0023]$
Increment, lower third	160	0.8924	0.9174	0.8255	0.0250	$[-0.0056, 0.0560]$
Increment, middle third	160	1.4411	1.4416	1.3768	0.0005	$[-0.0392, 0.0434]$
Increment, upper third	160	3.2522	3.2284	3.0924	-0.0238	$[-0.0804, 0.0271]$

Table 12: Raw primary energy score by year in standardized coordinates. Each row averages that year's two TEST months and the three diffusion seeds, using 24 test windows per year. Gaussian denotes the shared ensemble from Gaussian interpolation. These annual results use the existing fits and cases; each row covers only one year, so no interval from resampling years is reported.

Year	D	I-S	O-S	G5	Gaussian
2005	3.3402	3.4233	3.3349	3.1962	3.3435
2006	1.8130	1.8449	1.8911	1.8222	2.0005
2007	1.5185	1.6045	1.5749	1.4551	2.1359
2008	1.8584	1.8789	1.8882	1.7878	2.1347
2009	1.3782	1.4624	1.4206	1.2640	1.5893
2010	1.2774	1.3131	1.2785	1.2081	1.5417
2011	2.5543	2.5698	2.5144	2.3807	2.2021
2012	1.3628	1.3747	1.4155	1.3091	1.6150
2013	1.1809	1.2399	1.2585	1.1578	1.4643
2014	2.5162	2.4384	2.4978	2.4404	2.4855
2015	1.7493	1.7088	1.6719	1.6383	1.7181
2016	1.9325	1.8200	1.8618	1.7919	1.9331
2017	1.7522	1.7971	1.7414	1.7083	2.0006
2018	1.5171	1.5611	1.5522	1.4413	1.6696
2019	1.3378	1.3602	1.3326	1.3214	1.6536
2020	1.5016	1.5137	1.5409	1.4375	1.6264
2021	1.3806	1.4460	1.4136	1.3686	1.5898
2022	1.9837	2.0144	1.9358	1.9206	1.9962
2023	2.5198	2.3208	2.5221	2.1314	2.1605
2024	3.2793	2.9392	3.0514	2.9142	5.6425

F.1 Directional cancellation with thresholds from TRAIN

Directional cancellation occurs when the norm of the average vector is smaller than the average of the individual vector magnitudes. The cancellation analysis asks whether the identity penalty behaves differently in such cases. For each bin, define $C = (m^2 - \|\mu\|^2)/m^2$, with $C = 0$ when $m = 0$, and $A = (m^2 - \|\mu\|^2)/B_{\mathrm{ref}}^2$. Both are dimensionless. C measures the difference relative to that bin's squared mean magnitude; A instead uses the fixed TRAIN reference scale.

Group membership uses mean C over the 36 available bins or mean A over the twelve hidden bins. High cancellation means strictly above the TRAIN 75th percentile: 0.044395 for the available intervals

and 0.022696 for the missing interval. All 122,309 overlapping training windows contribute equally to these thresholds; normalization instead uses distinct bins. Although the rules were specified earlier, they were implemented after inspection of the primary results, so the analysis remains exploratory.

The energy groups use window means of Q/B_{ref}^2. Here $Q = m^2 + v_s$ is a magnetic second moment in nT^2; the ratio is dimensionless and is a different quantity from the statistical energy score. TRAIN quartile thresholds are $(0.349809, 0.571606, 1.078729)$ for the available intervals and $(0.343518, 0.563106, 1.069013)$ for the missing interval. A tie enters the higher energy group, whereas a tie at the cancellation threshold enters the rest group.

The analysis covers all 21 groups: the full test set, high cancellation and the rest, and combinations of cancellation with energy conditions. Group membership and scores for all 18 procedures are recorded. Each score gives equal weight to the months present, then averages the three diffusion seeds.

High cancellation in the available measurements identifies 128 windows across all twenty years. O-S minus D is -0.0256, with a 95% interval of $[-0.0740, 0.0147]$ from resampling years. All three seed differences are negative, but the interval still includes zero. O-S minus G5 is 0.0573, with interval $[0.0053, 0.1246]$, while O-S minus D is 0.0114 in the remaining 352 windows. These results do not establish an O-S advantage over both methods that learn from data alone. Grouping by the available context avoids selection on the hidden outcome, but it does not isolate cancellation from other correlated conditions.

High cancellation during the missing hour identifies 121 windows. O-S minus D is -0.0128, with interval $[-0.0858, 0.0629]$, compared with 0.0068 in the remaining 359 windows. Although TRAIN supplies the thresholds, the withheld outcome determines which TEST windows enter each group, so the comparison still depends on outcome selection.

The group combining high cancellation in the missing hour with the lowest energy quartile contains only seven windows from five years. Such comparisons describe losses on selected TEST cases. A claim of conditional predictive superiority would need an evaluation that accounts for the rule assigning cases to each group.

G Clipping draws to satisfy moment constraints and Gaussian regression

G.1 Algebraic completion and clipping of the existing draws

Two operations are applied separately to the same ensembles from the checkpoints selected by raw validation score. Model weights, draws and windows remain fixed. Algebraic completion derives v_v from the other five coordinates, leaving those coordinates and their primary energy score exactly unchanged. Training G5 also changes the generated representation and the loss, so its comparison includes more than algebraic completion alone.

Clipping can change the primary coordinates. The map retains μ, raises m to $r = \|\mu\|$ when necessary, and clips v_s to $[0, 1.5m^2 + 2.5r^2]$ using the updated m, before deriving v_v. The result satisfies the moment constraints and is realizable. Reapplying the map preserves it to the stated arithmetic tolerance. Minimum displacement is not established for this map.

For primary draws $X = (x_1, \ldots, x_K)$ and observation y, the empirical energy score obeys the displacement bound

$$|\mathrm{ES}(CX, y) - \mathrm{ES}(X, y)| \le \frac{2}{K} \sum_{i=1}^{K} \|Cx_i - x_i\|. \tag{8}$$

The reverse triangle inequality bounds the change in each score term by the mean displacement of the draws. Adding the two bounds gives Equation 8, without requiring a convex realizable set. The bound limits the magnitude of the score change; either sign is possible. Distances use all 60 primary coordinates after TRAIN standardization and are dimensionless. In Table 13, the change in distance to the withheld measurements is added and the change in the spread term is subtracted.

A deficit measures the size of an inequality violation in a draw before allowing for rounding. It is zero when the inequality holds. Failure frequency counts violations, regardless of their size. For G5, the 90th percentile of $\max(r - m, 0)/B_{\mathrm{ref}}$ is calculated within each window and averages to 0.01648, compared with 0.34407 for Gaussian interpolation. For $\max(-v_s, 0)/B_{\mathrm{ref}}^2$, the corresponding values are 0.000342 and 0.017372. Both deficits are dimensionless.

Table 13: Clipping the original ensembles at fixed model weights, draws and windows. Diffusion checkpoints are those selected by raw validation score. Months receive equal weight, followed by an average over three diffusion seeds or the one comparison ensemble. Distances use 60 primary coordinates standardized on TRAIN. Displacement is the mean distance moved by an ensemble member. The two signed term changes sum to the change in energy score before rounding. Training and validation choices remain fixed.

Procedure	Displacement	Δ distance	$-\Delta$ spread term	Δ energy score
D	0.1254	-0.00350	-0.00287	-0.00637
I-S	0.0997	-0.00427	-0.00119	-0.00547
O-S	0.1521	-0.01952	0.00080	-0.01873
G5	0.1055	0.00003	-0.00038	-0.00035
Gaussian interpolation	2.1403	-0.32056	0.42478	0.10422

Each percentile is calculated from the draws and hidden bins of one window, then averaged over months and seeds. This differs from a percentile of all samples pooled together. G5 and Gaussian interpolation both often violate the inequalities, but clipping moves their draws by different distances. G5's small score change is compatible with many raw draws lying outside the realizable set.

For D, clipping the ensemble from the checkpoint selected using raw draws changes the energy score by -0.00637 in Table 13. Selecting weights independently on VAL after clipping adds -0.01183; only seed 7 changes checkpoint. The difference between raw and clipped scores in Table 1 combines the effects of moving the draws and selecting the checkpoint.

G.2 Multivariate Gaussian regression

Gaussian regression predicts departures from interpolation using all 216 standardized available coordinates on both sides of the missing hour. Let X contain these coordinates and Y the 60 primary residuals from interpolation between bins 17 and 30, divided by the corresponding TRAIN scales. Fitting W and an unpenalized intercept b on TRAIN minimizes $\|Y - XW - b\|_F^2/N + \lambda\|W\|_F^2$, with equal weight for all 122,309 overlapping windows.

The nine candidates combine $\lambda \in \{0.001, 0.1, 1\}$ with covariance $(1-\rho)S + \rho\,\mathrm{diag}(S) + 10^{-6}I$, where $\rho \in \{0, 0.5, 1\}$ and S is the empirical residual covariance with divisor N. The conditional mean changes with context, while this residual covariance stays fixed. The Gaussian law covers the first five coordinates; deriving v_v produces a non-Gaussian distribution over all six. A positive-definite covariance defines a valid Gaussian law but does not enforce the moment constraints on its draws.

For each VAL window, all candidates use the same 39 innovations, generated with PCG64 and identified by split, window, draw and seed 13. Validation minimizes energy score with equal month weights, separately for raw and clipped draws, resolving exact ties in lexicographic (λ, ρ) order. Both procedures select $\lambda = 0.1$ and $\rho = 0$, with VAL scores 1.73275 and 1.83453. These choices were recorded before generating the single TEST ensemble.

Relative to Gaussian interpolation, regression changes the conditional mean fit and the finite noise realization as well as using more available measurements. Its score difference cannot isolate the benefit of the additional context alone.

G5 minus Gaussian regression has a mean difference in energy score of -0.1503, with year interval $[-0.2302, -0.0829]$ and interval from resampling groups of two years $[-0.2105, -0.0913]$. For O-S, the mean difference is -0.0502, with intervals $[-0.1336, 0.0327]$ and $[-0.0913, -0.0080]$, respectively. G5's intervals are negative under both schemes. O-S's interval includes zero when resampling years and excludes zero when resampling groups of two years. The grouping changes the resampling distribution and the uncertainty in this contrast.

These marginal 95% intervals use the original 10,000 temporal resamples and the same sampled years for both methods in each contrast. They hold the three diffusion fits and one Gaussian ensemble fixed and exclude variation from new fits, seed resampling and the validation choice. The later comparison uses a TEST period examined earlier.

Before clipping, Gaussian regression's vector energy score is 1.5728 in 36 standardized coordinates. Its CRPS for mean resolved magnetic energy density is 1.0497×10^{-12} J m^{-3}. These measure errors in different predictions on different scales. Clipping preserves the vector predictions but raises the CRPS for energy density to 2.1873×10^{-12} J m^{-3}. Mean displacement over the 60 standardized primary coordinates is 1.7733.

As with Gaussian interpolation, clipping Gaussian regression reduces the spread term more than the mean distance to the withheld measurements. Subtracting this smaller spread term raises the primary energy score even though the draws satisfy the moment constraints. Aggregate coverage near nominal and physical realizability do not by themselves establish an accurate predictive distribution.

H Penalty strength after training

Table 14 compares four penalty coefficients after 29,636 training steps: $c_R \in \{0, 0.00005, 0.0005, 0.005\}$, each with seeds 7, 13 and 42. Zero gives D; positive coefficients apply O-S's penalty on the variance identity. The final D and O-S models supply six fits, and the two additional coefficients require six more. Evaluation uses the same 480 windows, 39 draws and 50 DDIM evaluations as the main experiment. Samples before and after clipping use the same model weights and random draws. The final step was specified in advance. All twelve fits enter this exploratory comparison, without choosing a checkpoint or coefficient from their scores. Computation can differ despite equal step counts, and score differences may reflect convergence or overfitting.

Table 14: Penalty strength after 29,636 training steps. Each row gives one fit, with equal weight for the months present. Energy scores use the five primary quantities at twelve times in standardized coordinates. For samples before clipping, the remaining columns give mean absolute R_{obs} in nT^2, the percentage of bins failing a moment inequality beyond rounding, nominal 90% coverage, and mean interval width in standardized units. The inequality percentage excludes the equality test. All twelve fits are included, without using these scores to choose a checkpoint or coefficient.

	Energy score		Raw samples			
Coefficient / seed	Raw	Clipped	$\lvert R \rvert$	Bins failing inequalities (%)	Coverage (%)	Width
0 / 7	1.9787	1.9147	3.5152	78.4	57.5	0.8018
0 / 13	1.9778	1.9515	3.1092	68.1	67.9	0.8369
0 / 42	1.9900	1.9847	4.2064	11.4	60.6	0.7955
0.00005 / 7	2.0412	1.9714	3.5544	80.4	57.4	0.7830
0.00005 / 13	1.8735	1.8653	2.8988	59.7	69.4	0.8347
0.00005 / 42	1.9757	1.9702	3.8648	15.5	62.9	0.8396
0.0005 / 7	1.9558	1.9116	2.8245	88.8	60.4	0.7797
0.0005 / 13	1.8934	1.8824	2.3812	32.5	70.5	0.8112
0.0005 / 42	1.9312	1.9284	3.1843	15.6	66.8	0.8634
0.005 / 7	2.0224	1.9934	1.5068	69.1	57.2	0.7742
0.005 / 13	2.0201	2.0114	1.7024	54.4	61.4	0.8303
0.005 / 42	1.8119	1.8036	1.1243	44.2	68.6	0.7964

Figure 5 shows that the mean residual decreases as the penalty grows, while the mean energy score first improves and then worsens. The strongest penalty gives the smallest residual, while the original coefficient gives the best mean score. At $c_R = 0.005$, the score improves over D for seed 42 and worsens for seeds 7 and 13. A smaller identity residual can therefore accompany a worse predictive score.

At the original coefficient, O-S has a lower energy score than D for every seed at the end of training. When validation chooses the model weights, the difference changes sign across seeds. The penalty's apparent benefit therefore depends on the stage of training at which predictions are evaluated. Further fits would be needed to establish whether this pattern persists beyond the current exploratory comparison.

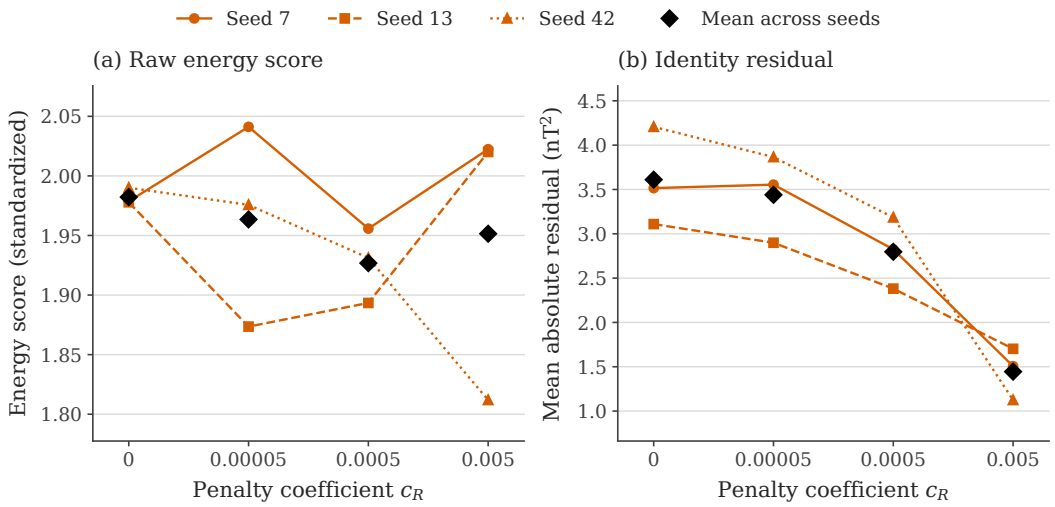


Figure 5: Energy score and identity residual after 29,636 training steps for the twelve models in Table 14. Lines connect the four coefficient settings for each seed; solid black diamonds mark the mean across three seeds. The settings are equally spaced on the horizontal axis. The left subplot shows raw energy score in standardized primary coordinates; the right shows mean absolute observation residual in nT^2. Each model value gives equal weight to the months present.

I Related work on physical constraints in prediction and generation

Gaussian process models estimate the solar wind at Mars between intermittent observations and quantify uncertainty [16]. Feedforward networks estimate second-order structure functions from incomplete magnetic records [1]. Diffusion models also predict the evolution of solar coronal images, with images as both conditioning data and predictions [17]. The solar wind experiment here models the joint distribution of magnetic statistics throughout a missing hour, conditional on measurements on both sides.

Bastek et al. [3, Section 3.3 and Appendix A.6.1] distinguish a residual at the denoiser mean from a residual at a sample, and demonstrate collapse when training uses the residual alone. The HMM here uses two evaluations of an exact denoiser to examine the sampling approximation separately from errors in learning the denoiser. The solar wind experiments compare stages of training and apply algebraic completion or clipping to the same samples. They assess effects on the predictive score that the residual alone cannot determine.

Physical emulators that produce one prediction for each input also use algebraic completion. Beucler et al. [5] compare training on all outputs, including those from conservation layers, with training only on direct outputs and completing the rest algebraically. G5 uses the latter construction in a conditional generator, changing both the generated representation and the training loss. Deriving v_v after sampling leaves the primary score unchanged. In climate downscaling, exact aggregation and nonnegativity impose distinct constraints, and predictive accuracy requires separate assessment [18].

PGDM downscales land surface temperature and compares arithmetic averaging with the fourth root of the mean fourth power of temperature [19]. The latter rule neglects variation in emissivity. PGDM conditions on geophysical variables, trains with MSE, and measures error in physical consistency during evaluation. WIND uses diffusion forcing for pretraining and moment matching for posterior guidance to reconstruct weather trajectories from observations that include spatial and temporal averages [20]. For guidance, the clean state conditional on the noisy state is approximated by a Gaussian. Its covariance is then propagated through a local linearization of the observation operator. Both methods place aggregation in the observation model, with training and inference that differ from the residual penalty examined here.

PILD models residuals with a Laplace likelihood whose scale adapts to their magnitude. For conditional tasks, it aligns representations with the output of an encoder whose weights remain

fixed during diffusion training [21]. To adjust the scale, it approximates conditional moments of residual magnitude from minibatch statistics with exponential averaging. Inference uses two denoiser evaluations, compared with 50 here. Physics-Based Flow Matching integrates the flow to $t = 1$ before evaluating residuals and combines gradients of the flow matching and residual losses through ConFIG [4]. The present experiment keeps c_R constant throughout training; adaptive objectives and gradient combinations fall outside its scope.

J Two examples of magnetic reconstruction

Figures 6 and 7 show reconstructions of two missing hours from TEST in physical units. Windows 120 and 360, counting from zero, are the midpoints of the two halves of the ordered list of 480 windows. The windows were chosen by position, independently of model error, after TEST had already been examined. Diffusion uses seed 7, the first declared seed, with model weights chosen from raw validation samples: training step 9560 for D and 29636 for O-S and G5. Gaussian regression uses the fit chosen by validation and the noise realization from seed 13. How representative these two missing hours are across TEST remains uncertain.

All methods receive the same measurements around the missing hour and are compared with the same withheld values. At each time and coordinate, the colored curve is the marginal median of the 39 samples. The nominal 90% interval includes the endpoints given by the second and thirty-eighth values in increasing order. Connecting the medians can produce an irregular curve that differs from every sampled trajectory. The curves and intervals describe each coordinate at each time separately; simultaneous coverage of a trajectory requires separate assessment. Thin lines show three sampled trajectories, ensemble members 0, 1 and 2.

Times mark the starts of five-minute bins relative to the missing hour. Predictions are shown before clipping. Figures 6 and 7 use a common vertical scale for each quantity across methods. The symmetric logarithmic scale for v_s displays small measured variances and Gaussian predictions outside the physical range. The task reconstructs statistics over five minutes; different sequences of minute averages can produce the same statistics.

The sampler keeps the available measurements unchanged. The plots draw these measurements and the twelve predicted bins as separate segments, leaving short breaks from -5 to 0 minutes and from 55 to 60 minutes. Neighboring bins contain different sets of five vectors. The variance identity applies inside each bin, while the conditional model learns dependence across time. The elevated variance predictions are model errors, visible on the linear scales in Figure 8.

Each bin contains five vector averages, each over one minute; v_s is the variance of their magnitudes. In both examples, diffusion medians generally exceed the withheld variance and the nominal intervals often miss it. In TEST window 120, each diffusion method covers v_s in zero of the twelve hidden bins. In window 360, D, O-S and G5 cover zero, one and two bins, respectively. Gaussian regression covers all twelve bins in both windows with much wider intervals. It uses the same residual covariance for every context, and its distribution assigns probability to negative v_s, which is physically impossible. The aggregate results also show undercoverage of the variance. How often the particular errors in these examples occur across TEST, and how they arise during training, remain uncertain.

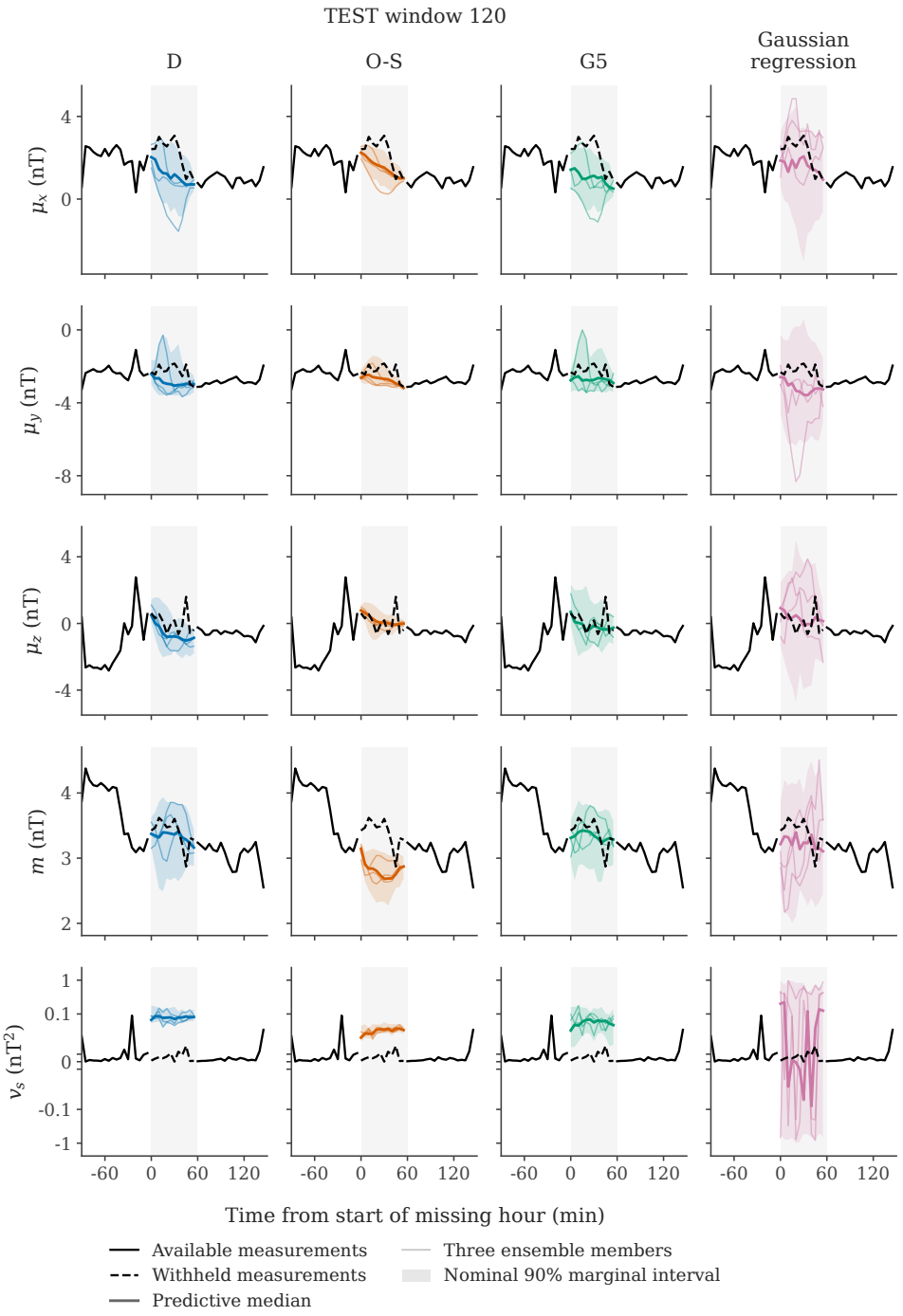


Figure 6: Reconstruction of TEST window 120 by D, O-S, G5 and Gaussian regression (columns), with model weights chosen by validation. Rows show the five primary magnetic statistics in physical units. All samples are shown before clipping; diffusion uses seed 7 and Gaussian regression uses seed 13. Solid black lines show the 90 minutes of available observations on each side of the shaded missing hour; dashed black lines inside it show withheld measurements. Colored lines give marginal medians, bands give nominal 90% intervals, and thin lines show ensemble members 0, 1 and 2. Times mark the starts of five-minute bins relative to the start of the missing hour. The v_s axis is linear within 0.05 nT^2 of zero and logarithmic outside, so equal vertical distances can represent different absolute errors in that row. Negative variance predictions remain visible. Appendix J explains the window choices and interval construction.

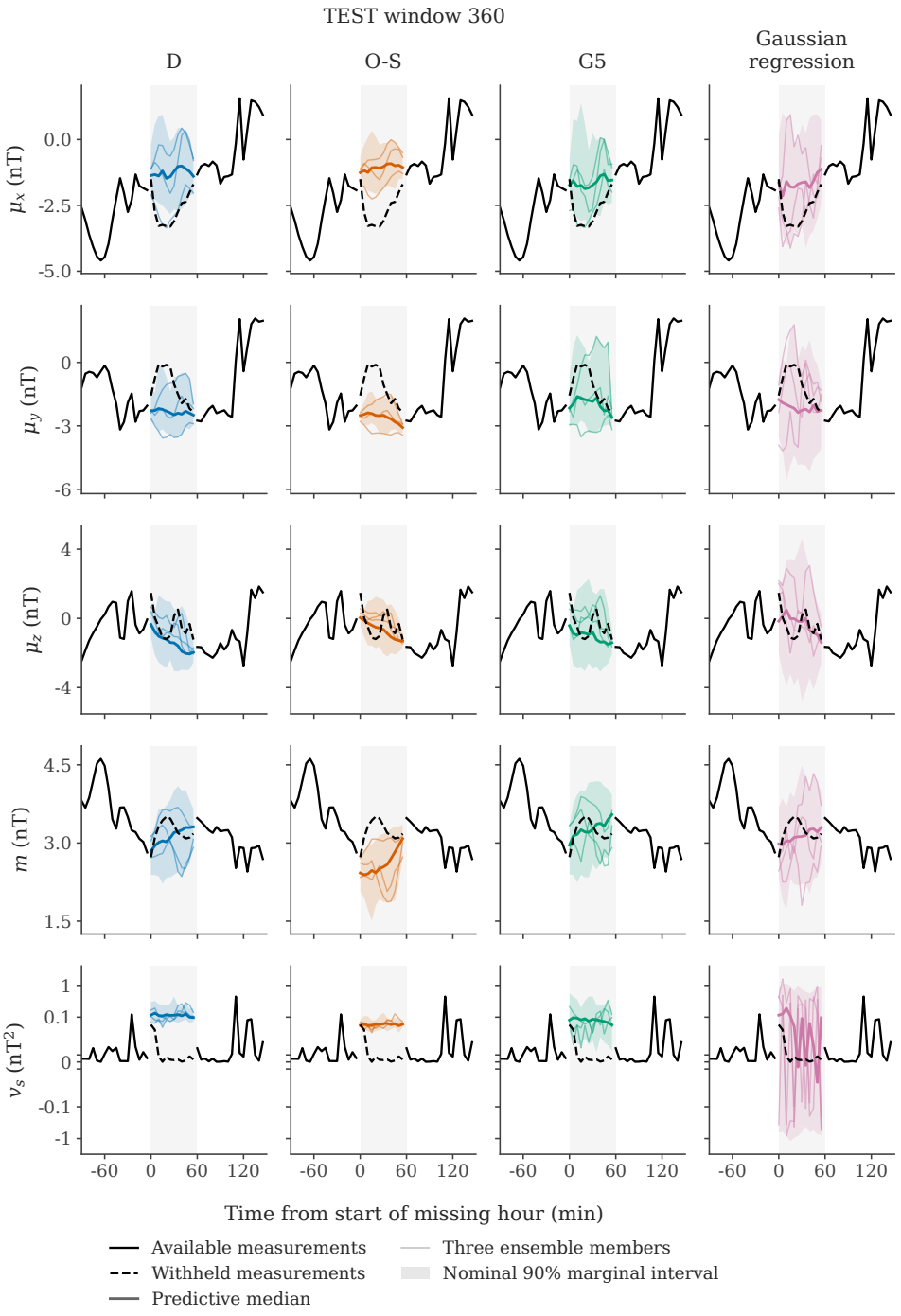


Figure 7: Reconstruction of TEST window 360, using the same methods, seed choices, units and plotting conventions as Figure 6. Each predictive interval is calculated from the ensemble for its method.

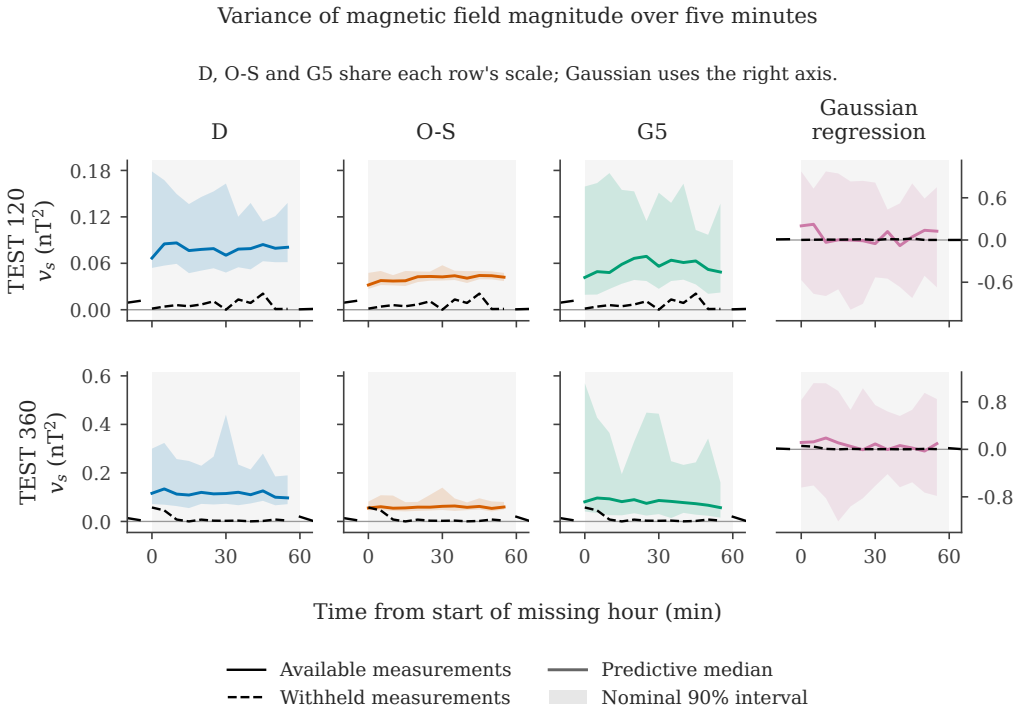


Figure 8: Variance of magnetic field magnitude within five-minute bins during the same two missing hours. D, O-S and G5 share a linear scale within each row; Gaussian regression uses a separate linear scale on the right. Vertical distances and band heights are comparable only on the same scale. The axis limits include all measurements, medians and interval ends in each subplot. Solid black lines show available observations; dashed black lines show withheld measurements. Colored lines and bands give empirical medians and nominal 90% marginal predictive intervals from all 39 samples. Times mark bin starts relative to the start of the missing hour. The horizontal gray line marks zero; predictions below it violate the nonnegativity of variance. The Gaussian subplots show much wider intervals that contain the observations.